

Phase 1 Experiment Analysis

Overview

Phase 1 evaluates **5 key builders** x **8 weight configurations** = 40 runs on `libero_spatial` (10 tasks, 5 episodes each). The search strategy is `weighted_rrf_knn` across all runs; only the key builder type and field weight allocation vary.

Key Builders

ID	Type	Description
a	<code>cp1_mean_pool</code>	Global average pooling of intermediate features
b1	<code>cp1_spatial_pool_16</code>	Spatial pooling to 16-dim
b2	<code>cp1_spatial_pool_64</code>	Spatial pooling to 64-dim
c	<code>cp1_max_pool</code>	Global max pooling of intermediate features
d	<code>clip</code>	CLIP visual encoder features

Weight Configurations

All weights are over `(vision_0, vision_1, prompt_emb, robot_state)`. `vision_2` is disabled in all runs. `prompt_emb` is always 0.

ID	vision_0	vision_1	robot_state	Design Intent
w1	1.0	0	0	Vision-only baseline
w2	0	0	1.0	Robot-state-only baseline
w3	0.5	0	0.5	Equal v0/rs split
w4	0.25	0	0.75	RS-heavy, single vision
w5	0.25	0.25	0.5	Dual vision + RS
w6	0.15	0.1	0.75	RS-heavy, dual vision (low)
w7	0.1	0.1	0.8	RS-dominant, minimal vision
w8	0.5	0.25	0.25	Vision-heavy mix

Results

Full Results Table (Success Rate)

	w1	w2	w3	w4	w5	w6	w7	w8	Mean	Max	Min	Std
mean_pool	26.0	55.6	42.0	50.0	58.0	52.0	56.0	52.0	48.9%	58.0%	26.0%	9.8%
spatial_16	36.0	46.0	44.0	58.0	58.0	56.0	58.0	62.0	52.2%	62.0%	36.0%	8.5%
spatial_64	32.0	46.0	42.0	40.0	64.0	50.0	58.0	60.0	49.0%	64.0%	32.0%	10.3%
max_pool	40.0	46.0	42.0	56.0	50.0	52.0	54.0	48.0	48.5%	56.0%	40.0%	5.3%
clip	12.0	46.0	44.0	50.0	60.0	66.0	66.0	52.0	49.5%	66.0%	12.0%	16.2%

Per-Weight Best Key Builder

Weight	Best Key Builder	Success Rate
w1	max_pool	40.0%
w2	mean_pool	55.6%
w3	spatial_16	44.0%
w4	spatial_16	58.0%
w5	spatial_64	64.0%
w6	clip	66.0%
w7	clip	66.0%
w8	spatial_16	62.0%

Analysis

1. Weight Allocation Findings

Vision-only retrieval (w1) performs worst across all key builders. Every key builder achieves its lowest success rate with w1. This indicates that a single vision feature alone is insufficient for effective cache retrieval in LIBERO spatial tasks.

Higher robot_state weight generally yields better results. Configurations w2 (rs=1.0), w4 (rs=0.75), w6 (rs=0.75), and w7 (rs=0.8) consistently rank among the top performers. In LIBERO spatial tasks, robot state contributes more to cache retrieval quality than visual features.

Multi-modal fusion outperforms single-modality retrieval. Mixed-weight configurations (w3–w8) generally outperform single-feature baselines (w1, w2). The best overall results come from w6 (v0=0.15, v1=0.1, rs=0.75) and w7 (v0=0.1, v1=0.1, rs=0.8), both achieving 66% with clip — suggesting that a small amount of visual signal on top of a dominant robot_state weight is the optimal regime.

Weight allocation matters more than key builder choice. Within the same key builder, success rates vary by 20–54 percentage points across weights. Across key builders at the same weight, variation is typically under 15 points. This means tuning the weight vector is higher-leverage than switching feature extractors.

2. Key Builder Comparison

Ranking by mean success rate:

Rank	Key Builder	Mean	Max	Min	Std
1	spatial_16	52.2%	62.0%	36.0%	8.5%
2	clip	49.5%	66.0%	12.0%	16.2%
3	spatial_64	49.0%	64.0%	32.0%	10.3%
4	mean_pool	48.9%	58.0%	26.0%	9.8%
5	max_pool	48.5%	56.0%	40.0%	5.3%

spatial_16 is the best all-rounder. It achieves the highest mean success rate (52.2%) with moderate variance (std=8.5%). It wins 3 out of 8 per-weight comparisons (w3, w4, w8) and is consistently competitive across all weight configurations.

clip has the highest ceiling but the lowest floor. It achieves the global best of 66% (w6 and w7) but collapses to 12% with pure vision retrieval (w1). Its std of 16.2% is nearly double the next highest. clip is extremely sensitive to weight allocation — it requires high robot_state weight to be effective, but when properly configured, it outperforms all other key builders. Additionally, **using clip increases VRAM usage by nearly 3 GB** due to loading the CLIP model, and **this overhead is incompressible.**

max_pool is the most stable but has a low ceiling. With a std of only 5.3%, it delivers consistent results (40–56%) regardless of weight choice. However, its maximum of 56% is the lowest among all key builders. It is a safe but unexciting choice.

spatial_64 does not improve over spatial_16. Despite encoding 4x more spatial information, spatial_64 achieves a lower mean (49.0% vs 52.2%) with higher variance (10.3% vs 8.5%). Higher spatial resolution does not help in this task setting.

mean_pool is unremarkable. It ranks second-to-last by mean. Its one standout result — highest w2 score (55.6%) — suggests its visual features are weak, and performance depends primarily on robot_state retrieval.

Recommendations for Phase 2

1. **Focus weight search on the robot_state-dominant regime** ($r_s = 0.7\text{--}0.9$), with small vision contributions. The w6/w7 region appears most promising.
2. **Prioritize spatial_16 and clip** as key builder candidates. spatial_16 for stability, clip for peak performance.
3. **Consider dropping spatial_64 and max_pool** — spatial_64 offers no advantage over spatial_16, and max_pool has a low ceiling.
4. **Investigate whether clip + spatial_16 ensemble** could combine clip's high ceiling with spatial_16's stability.

Limitations

The current experiment uses only 5 episodes per task across 10 tasks, resulting in a small sample size. Success rate differences of a few percentage points may not be statistically significant at this scale. However, running broader experiments (more episodes, more task suites) is prohibitively slow at this stage. The purpose of Phase 1 is to narrow down the search space. Once a final configuration is selected, a larger-scale validation experiment should be conducted to confirm the results with higher statistical confidence.